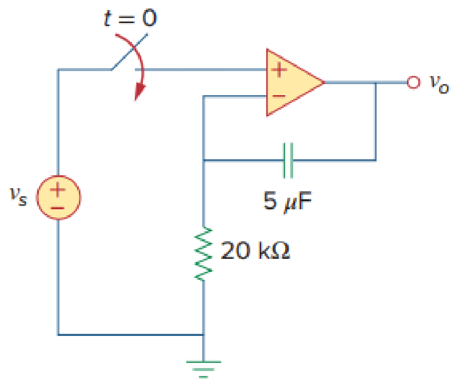
## Problem

Determine *vo* for *t* > 0 when *vs* = 20 mV in the op amp circuit of Fig. 7.135.

**Fig. 7.135:**




## Step-by-step solution

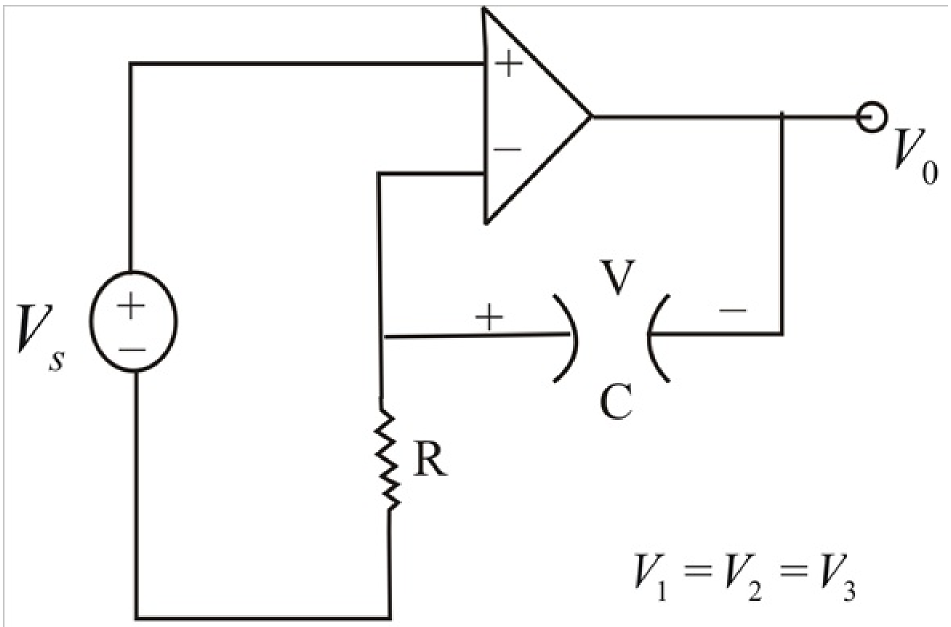
**Step 1** of 4

Let V = Capacitor voltage.

For t < 0, Switch is open & V(0) = 0.

For t > 0, Switch is closed and the circuit becomes as shown below.

**Step 2** of 4




$$V_1 = V_2 = V_3$$

**Step 3** of 4

$$V_1 = V_2 = V_S \rightarrow (1)$$

$$\frac{0-V_S}{R} = C\frac{dV}{dt} \rightarrow (2)$$

Where,

$$V = V_S - V_0 \Rightarrow V_0 = V_S - V \rightarrow (3)$$

From(1)

$$\frac{dV}{dt} + \frac{V_S}{R_C} = 0$$

$$V = -\frac{1}{R_C}\int V_S \; dt + V(0) = \frac{-tV_S}{R_C}$$

**Step 4** of 4

Since $V_S$ is constant.

$$R_C = (20\times10^3)\,(5\times10^{-6}) = 0.1$$

$$V = \frac{-20}{0.1}mV = 200t\,mV$$

From(3),

$$V_0 = V_S - V = 20 + 200t$$

$$= 20(1+10t)mV \; .$$